

Shot Selection Efficiency in Basketball: A Revealed Efficiency Approach

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Background

- We create a shot selection measure called *Shot Selection Efficiency* that takes as input a team's (player's) expected points and expected proportional volume from each observed shooting bin of the half court and computes the team's bin shot volume weighted correlation between bin-level proportional volume and bin-level expected points, from the field and free throw line, across all shooting bins for a season.

Methods

- Play-by-play data was subscribed from bigdataball.com for the 2016/17-2022/23 seasons.

- Binned shots and computed expected true points:

$$TP_{ijk} = \frac{\text{makes}_{ijk} \cdot \text{shot.type}_j + \text{foul.shots.resulting}_{ijk} \cdot \text{FT}\%_{ik}}{\text{shot.attempts}_{ijk}}$$

- Shot Selection Efficiency* was then calculated as the shot-volume weighted correlation between bin-level expected true points and bin-level shooting volume across all shooting bins for a season.
- Exogenous Shooting Ability* is obtained by finding a team's average expected true points from each bin of its shot chart in a season. Then, we take an unweighted average of all bin-level expected true point values to obtain a measure of shooting ability that is exogenous of shot selection.
- We first run 2 sets of 4 fixed effects regression models. On LHS, we specify either team-season *Pythagorean expected win%* or *team-season actual win%*..
- On the RHS, we essentially specify the four factors on each side of the ball, but at different levels of aggregation or disaggregation, toward highly controlled, highly-explanatory models that isolate the effect of shot selection efficiency on team success.
- Explanatory team-season variables: *DRtg*, *exogenous shooting ability*, *shot selection efficiency*, *ORB%*, *TOV%*, *team fixed effects*, and *season*.
- A third set of models consider the payroll-conditional effect of *Shot Selection Efficiency* on team success. Is *Shot Selection Efficiency* priced-in or Moneyball?

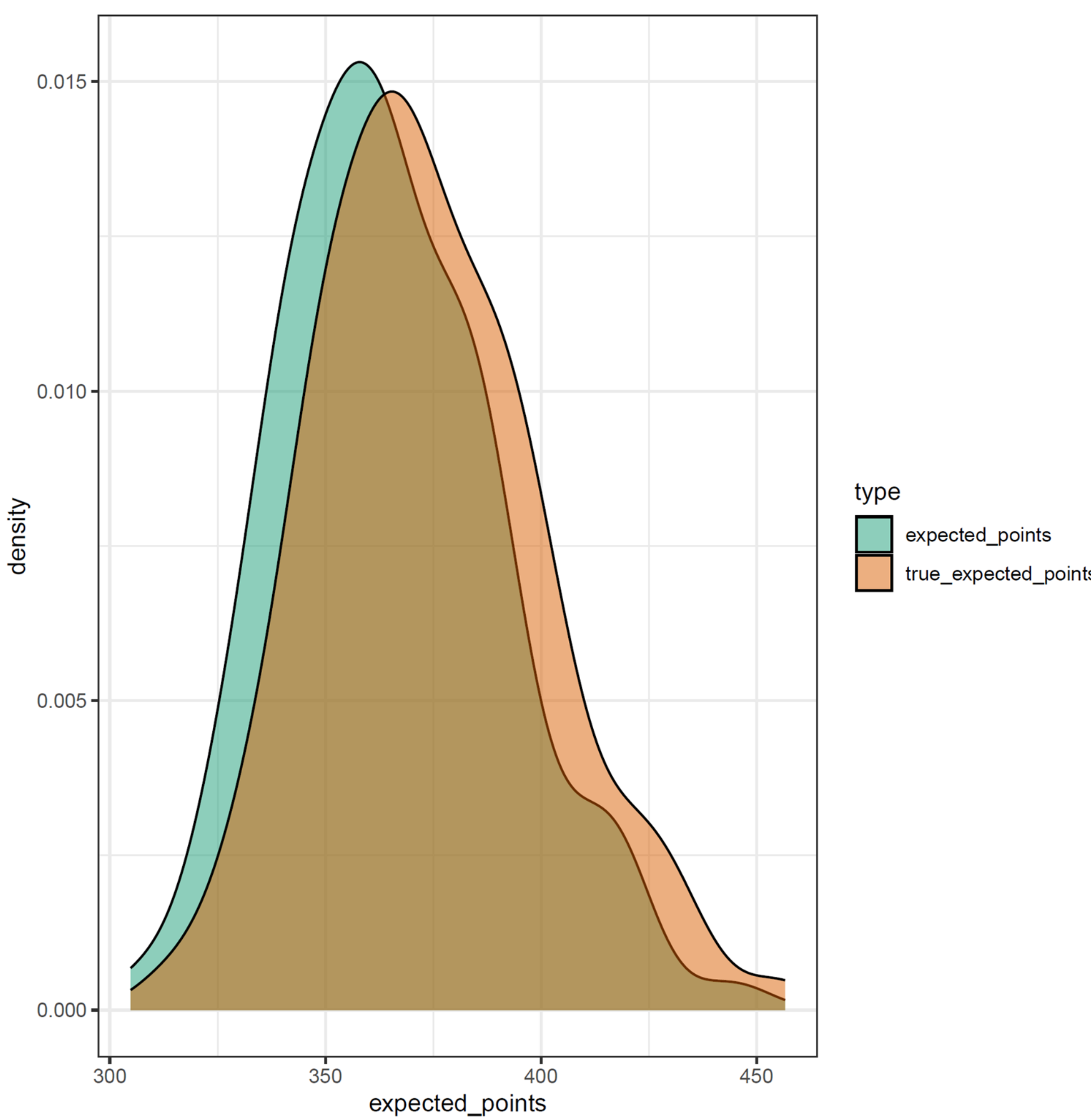
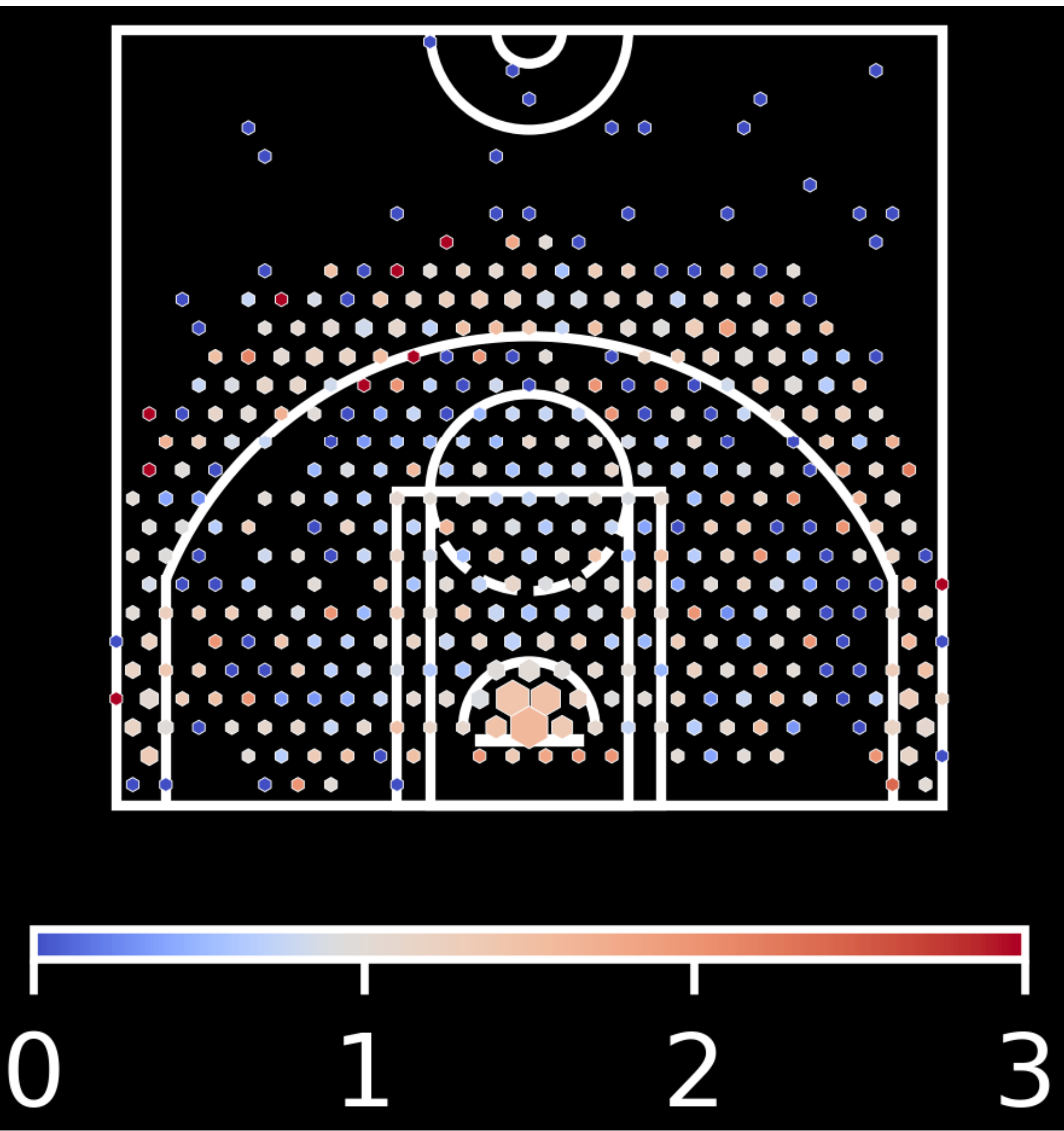
Characteristic	N = 210
Pyth_W_pct	
Median (SD) Range	0.50 (0.14) 0.18 - 0.82
W_pct	
Median (SD) Range	0.50 (0.14) 0.18 - 0.82
NRtg	
Median (SD) Range	0.4 (4.6) -10.5 - 11.6
true_shooting_efficiency	
Median (SD) Range	0.51 (0.05) 0.40 - 0.61
exogenous_shooting_ability	
Median (SD) Range	0.88 (0.05) 0.76 - 1.03
off_TOV%	
Median (SD) Range	12.60 (0.90) 9.90 - 14.90
off_ORB%	
Median (SD) Range	22.80 (2.21) 17.90 - 30.20
DRtg	
Median (SD) Range	111.2 (3.3) 102.9 - 120.0
payroll_normalized	
Median (SD) Range	0.78 (0.11) 0.46 - 1.00

Results

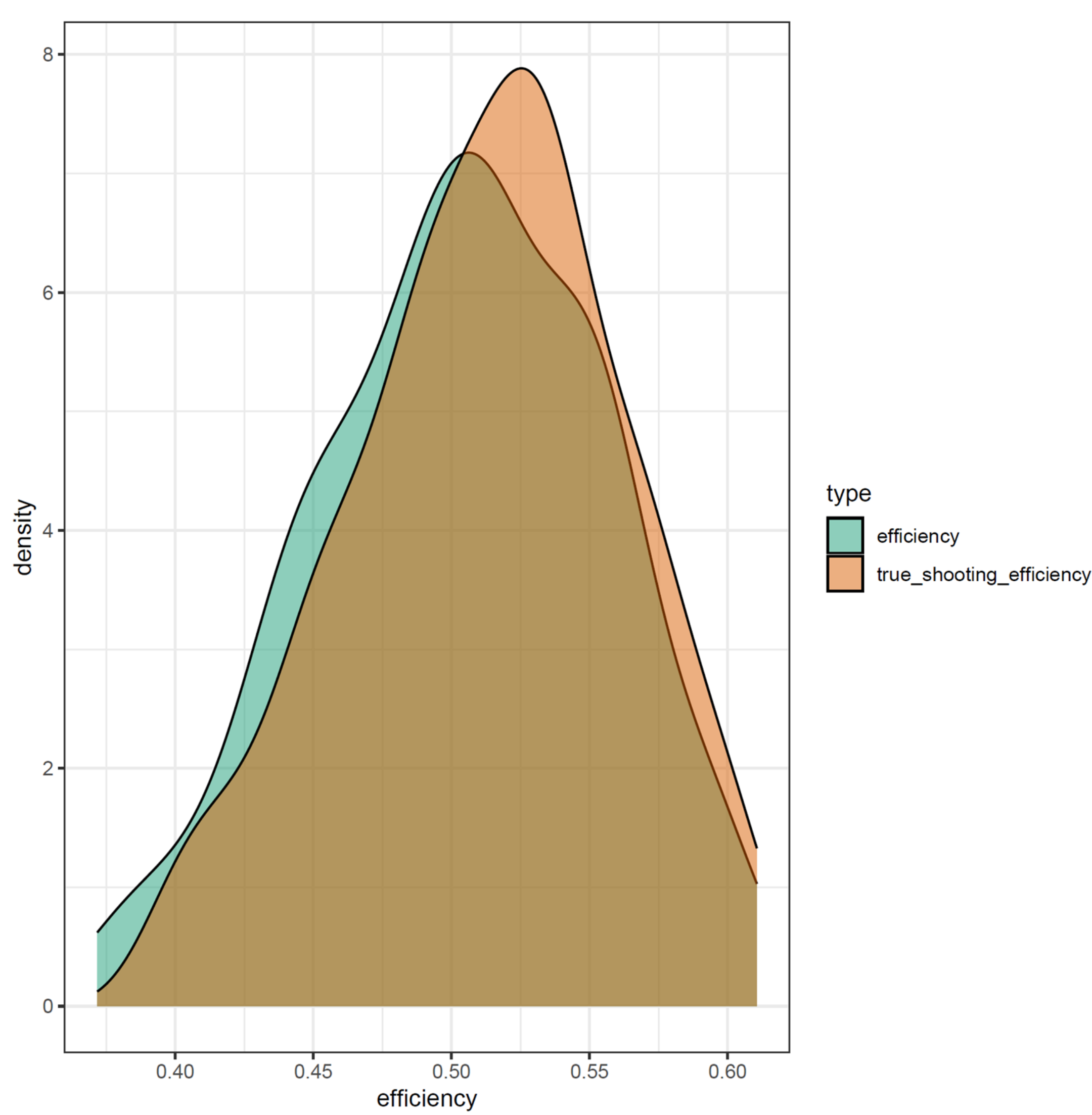
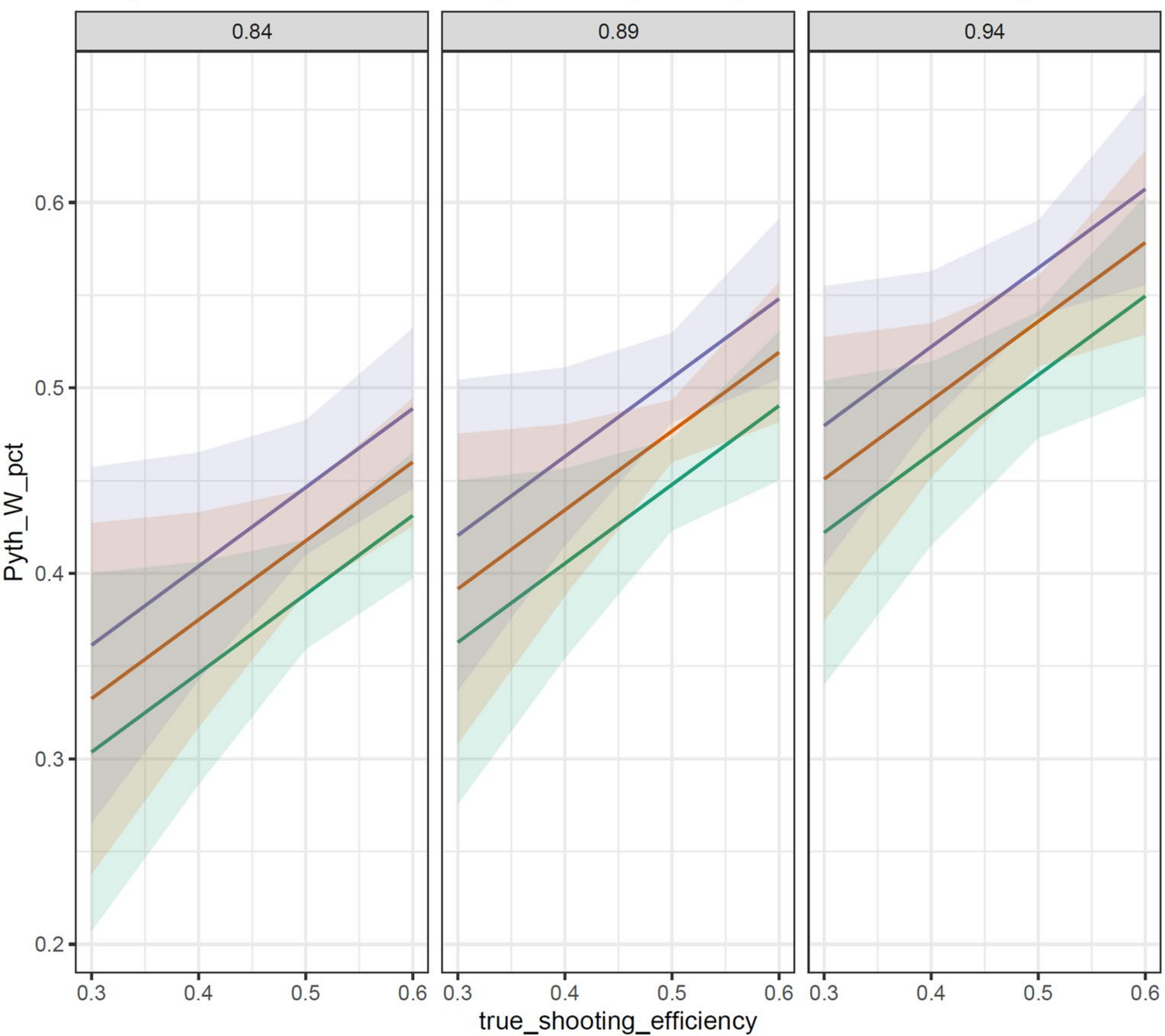
Estimated Models *All models include team fixed effects.

Predictors (Intercept)	Pyth_W_pct			W_pct			Pyth_W_pct		
	Estimates	std. Error	p	Estimates	std. Error	p	Estimates	std. Error	p
DRtg	-0.031	0.002	<0.001	-0.031	0.002	<0.001	-0.031	0.002	<0.001
true shooting efficiency	0.350	0.118	0.003	0.295	0.136	0.032	0.333	0.148	0.025
exogenous shooting ability	1.050	0.127	<0.001	1.062	0.146	<0.001	1.095	0.150	<0.001
season	0.017	0.003	<0.001	0.016	0.004	<0.001	0.022	0.004	<0.001
off TOV%	-0.036	0.006	<0.001	-0.035	0.007	<0.001			
off ORB%	0.016	0.002	<0.001	0.013	0.003	<0.001			
payroll normalized							1.095	0.619	0.079
payroll normalized^2							-0.674	0.388	0.084
Observations	210			210			210		
R ² / R ² adjusted	0.827 / 0.792			0.782 / 0.738			0.739 / 0.687		

2021-22 Milwaukee Bucks True Shot Chart



Marginal Effect of True Shooting Efficiency on Expected Win Percentage



Conclusion

- A new shot selection measure, called *Shot Selection Efficiency*, aggregates from team-season shot charts into a single efficiency measure that incorporates shot-dependent FT scoring.
- The measure does not depend on overall shooting ability. A separate measure of exogenous shooting ability is created that is exogenous of shot selection.
- In 2-way, 4-factor type models, *Shot Selection Eff.* and *team success* have a positive, significant rel'n.
- If a team in a season can improve the weighted correlation between bin-level shot-volume and expected true points by 0.1 units, it gains an expected 0.035 units of win proportion or nearly 3 games. Alternatively, a 1 s.d. increase in the variable increases expected wins by 1.41 according to the full Pyth_W_pct model.
- A 1 s.d. increase in *Exogenous Shooting Ability* has approximately a three-fold effect on expected wins, by comparison.
- A third set of models suggests *Shot Selection Efficiency* is a source of expected wins not fully explained by team payroll.
- Shot Selection Efficiency* may not be fully priced into the player market, suggesting a potential Moneyball opportunity related to game/set management of shot selection.
- Future work will examine whether *Shot Selection Efficiency* is a measure of team coaching efficiency via coaching tenure fixed effects.

